

Does semantic valence induce a halo effect on facial attractiveness ? A Diffusion Decision Model approach

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ABSTRACT

Facial attractiveness strongly impacts social decisions, often leading to the 'halo effect' where attractive people are assumed to have positive traits. In the context of dating apps, where users make split-second binary choices, we investigate how semantic information (personal descriptions) interacts with visual attractiveness. Specifically, does a positive description make a face seem more attractive? Using the Diffusion Decision Model (DDM), we analyzed reaction times and choices to understand the cognitive mechanism: does semantic context bias our initial readiness to decide (starting point), or does it change how fast we process the visual evidence (drift rate) ?

Keywords: Attractiveness, DDM, Halo Effect

1 INTRODUCTION

The “what is beautiful is good” stereotype refers to the tendency to attribute more positive personal characteristics to physically attractive individuals compared to less attractive ones [Dion, Berscheid, & Walster, 1972]. This phenomenon is commonly explained by the *halo effect*, a cognitive bias whereby the evaluation of one salient attribute influences judgments of other unrelated traits [Nicolau et al.]. Cognitive biases are mental shortcuts that help individuals process information efficiently, but they can also lead to systematic errors in judgment [Little and Wu].

Facial attractiveness plays an important role in social decision-making, especially in situations where judgments are made quickly and with limited information. This is particularly the case in dating applications, where users often rely on a small number of pictures to decide whether they like a person or not. These decisions are typically rapid and binary, and are largely based on facial appearance, making dating applications a relevant context for studying biased decision-making.

Beyond pictures, dating applications also display short biographies or tags provide that allow users to share information about themselves. This raises the question of whether such semantic information can influence judgments of facial attractiveness. In other words, can positive personal information make a face appear more attractive ? Or can a glowing personality description override a neutral face, or boost an already attractive one ? Previous study suggest that facial attractiveness is not determined solely by facial features, but can be influenced by contextual, semantic and moral cues [Zhang et al., Luo et al.]. Zhang’s team showed that presenting two-word personality traits before

a female face significantly altered how its attractiveness was rated. When positive personality traits (e.g., decent, honest) were presented before faces, participants assigned higher attractiveness ratings, whereas negative traits (e.g., evil, mean) presented beforehand significantly reduced these ratings compared with no description at all. Similarly, in Luo’s study, participants viewed faces paired with short sentences describing positive, neutral, or negative behaviour before rating their attractiveness. When a negative behaviour (e.g., “He betrayed friends for money”) accompanied a face, participants’ attractiveness ratings decreased, whereas positive behaviour (e.g., “He saved a drowning child”) increased these ratings compared with the neutral behaviour (e.g., “He chatted with classmates”). These findings provide a good example of the *halo effect*.

To understand the cognitive mechanisms driving these rapid attractiveness judgments, we applied the Diffusion Decision Model (DDM). This model translates “behavioral data—accuracy, mean response times, and response time distributions—into components of cognitive processing” in two-choice decision tasks [Ratcliff and McKoon]. In the present study, this framework is used to examine whether semantic valence influences facial attractiveness judgments by biasing the decision process.

The DDM decomposes decision-making into three main parameters. The starting point (z) reflects a pre-decisional bias toward one of the two response options (attractive vs. unattractive). The drift rate (v) represents the speed and direction of evidence accumulation based on facial information, which may be biased by the preceding semantic information. Finally, the boundary separation (a) reflects the amount of information required to make a decision and is related to the speed-accuracy trade-off. Because the task instructions were the same for all trials, we do not make specific predictions about changes in this parameter.

Together, these parameters allow us to test whether the semantic halo effect arises from early biases in decision onset (starting point) or from biased interpretation of facial information during evidence accumulation (drift rate).

2 LITERATURE REVIEW

2.1 Facial Attractiveness

Facial attractiveness refers to an evaluative judgment of a person’s face that is formed rapidly and shows a high degree of agreement across observers. In a meta-analytic

and theoretical review, [Langlois et al.] demonstrated that judgments of facial attractiveness are highly consistent across individuals and cultures, and that attractive individuals are systematically evaluated more positively on a wide range of social traits, such as competence, intelligence, and sociability. These findings suggest that facial attractiveness is a robust and socially meaningful cue rather than a purely subjective preference.

From a social-cognitive perspective, facial attractiveness is not limited to the perception of physical features. Instead, it reflects an integrative evaluation that combines perceptual information with higher-level social interpretations. [Zebrowitz and Montepare] argued that facial appearance plays a central role in social perception because it provides rapid information that is used to form impressions about others. Similarly, Todorov et al. (2008) showed that faces are spontaneously evaluated along socially relevant dimensions, such as trustworthiness or dominance, even when observers are not explicitly instructed to make such judgments. This suggests that facial evaluations are closely linked to automatic social inference processes.

Importantly, facial attractiveness has strong consequences for social evaluation. A large body of research indicates that attractive individuals are generally judged more favorably on multiple personal characteristics, a phenomenon commonly referred to as the “what is beautiful is good” stereotype. These positive attributions extend beyond aesthetic judgments and influence broader social perceptions and expectations.

Moreover, facial attractiveness is not a fixed or purely visual property. Contextual and semantic information associated with a person can modulate how attractive a face is perceived. For example, [Zhang et al.] showed that faces preceded by positive personality descriptions were judged as more attractive than the same faces preceded by neutral descriptions. This finding indicates that information about personal qualities can influence aesthetic judgments, highlighting the role of higher-level cognitive processes in facial attractiveness perception.

These studies suggest that facial attractiveness operates as a rapid and that integrates evaluation, which we believe makes it particularly relevant for studying halo effects in fast social decisions. As a result, facial attractiveness can act as a gateway for broader social inferences, making it a key mechanism through which halo effects may emerge in social evaluations.

2.2 Halo Effect

As introduced earlier, the halo effect refers to a cognitive bias whereby the evaluation of one salient attribute influences judgments of other, unrelated attributes. This concept was originally introduced by [Thorndike], who observed that evaluators tend to form a global impression of an individual, which then biases the assessment of specific characteristics. As a result, different attributes are not evaluated independently, but rather through the lens of an overall positive or negative impression.

In the context of social perception, the halo effect has been widely used to explain why attractive individuals are often perceived more favorably on a range of social traits. This phenomenon is closely related to the “what is beautiful is good” stereotype, according to which physical attractiveness leads to more positive judgments of competence, intelligence, or trustworthiness (Dion et al., 1972). Facial attractiveness can therefore act as an initial cue that shapes broader social evaluations, even when these traits are not directly observable.

More recent work has provided empirical evidence that the halo effect reflects a genuine bias in evaluative processes rather than a mere correlation between attributes. Using a large longitudinal dataset of online hotel reviews, Nicolau et al. (2020) formally tested the halo effect by examining how changes in the evaluation of certain attributes influenced the evaluation of other, attributes that should be stable. Their results showed that nearly half of the variance in the evaluation of a focal attribute could be explained by changes in other attributes, providing strong evidence for the existence of a halo effect. Importantly, they also reported asymmetric effects, with negative information exerting a stronger influence than positive information, a phenomenon referred to as the *horn effect*.

The authors focused on consumer evaluations rather than social perception per se, their findings highlight a general cognitive mechanism : individuals do not evaluate attributes in isolation, but rather rely on global impressions that bias subsequent judgments. This mechanism is highly relevant to social evaluations of faces, where an initial impression based on attractiveness may influence judgments of unrelated traits such as trustworthiness.

Taken together, these findings suggest that the halo effect operates as a systematic bias in evaluative judgments, driven by initial impressions that spill over onto other dimensions. However, while the halo effect is well established at the behavioral level, less is known about the cognitive processes through which such biases emerge over time. This motivates the use of computational models, such as the Diffusion Decision Model, to disentangle the mechanisms underlying halo effects in rapid social decision-making.

2.3 Decision Diffusion Model (DDM)

Although the halo effect has been extensively documented at the behavioral level, traditional analyses based on accuracy or mean reaction times do not allow a precise identification of the cognitive mechanisms underlying this bias. In particular, such measures make it difficult to determine whether halo effects originate from pre-decisional tendencies or from biased information processing during the decision itself.

The Diffusion Decision Model (DDM) provides a formal framework to address this limitation by modeling two-choice decisions as a process of noisy evidence accumulation toward one of two response boundaries [Ratcliff and McKoon]. Within this framework, the starting point (z) reflects pre-decisional bias toward a given response, whereas the drift rate (v) reflects the quality and direction of evidence accumulation. Additional parameters, such as boundary

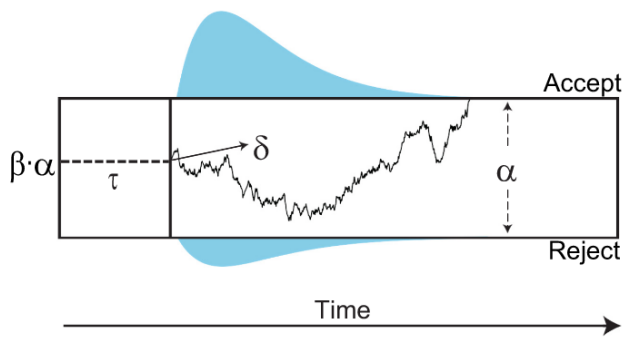


Fig. 1. Decision diffusion model (DDM). Participants begin with an initial bias, β , reflecting a predisposition to judge the facial attractiveness of the presented face. Noisy evidence is accumulated over time, with δ indicating the direction of evidence accumulation toward one of the two response options. The parameter α represents the boundary separation, corresponding to the amount of evidence required to make a decision. The parameter θ' reflects non-decision processes (e.g., perceptual encoding and motor execution). The model predicts the distribution of response times for accept (here, attractive) and reject (here, unattractive) decisions (shown in blue) [Axt and Johnson].

separation (α) and non-decision time (τ), capture decision caution and peripheral processes, respectively.

Importantly, recent work has shown that social biases can be decomposed into these distinct DDM components. In a series of studies, [Axt and Johnson] applied the DDM to social and evaluative judgments in order to investigate the mechanisms underlying discrimination and person perception biases. Their results shows that socially relevant cues can bias both the starting point of the decision process and the drift rate, indicating that such biases operate at multiple stages of decision-making. Specifically, individuals showed initial response biases favoring socially valued categories, as well as faster evidence accumulation toward congruent judgments.

These findings are particularly relevant for the study of halo effects in facial evaluation. If facial attractiveness or positive semantic information biases the starting point, this would suggest that halo effects partly reflect an a priori preference toward positive judgments. Alternatively, if such information modulates the drift rate, this would indicate that incoming information is processed more efficiently or interpreted more positively for attractive individuals. The DDM thus provides a principled way to dissociate these mechanisms and to determine whether halo effects reflect pre-decisional biases, biased evidence accumulation, or a combination of both.

In the present study, the DDM is used to investigate how semantic valence interacts with facial attractiveness in shaping judgments of attractiveness. By decomposing decisions into distinct cognitive components, this approach allows for a mechanistic account of halo effects in rapid social decision-making tasks.

3 AIMS AND GOALS

The objective of this study is to investigate whether semantic valence influences judgments of facial attractiveness and to examine the cognitive mechanisms underlying the *halo effect* using the Diffusion Decision Model (DDM). More specifically, this study addresses the following research questions:

RQ1: Does the valence of semantic information presented prior to viewing a human face influence reaction times in a decision-making task?

RQ2: Does semantic valence affect the decision-making process by biasing the starting point of evidence accumulation, the drift rate, or both, as described by the DDM?

3.1 Hypotheses

Participants performed a speeded binary decision task (attractive/unattractive), indicating whether each face was attractive or not. On each trial, a word with positive, negative, or neutral semantic valence was first presented, followed by a face stimulus that had been pre-classified as attractive or unattractive based on a standardized database. Each face appeared only once during the experiment. Based on previous research on semantic priming, facial attractiveness, and halo effects, the following hypotheses were formulated.

H1 (Reaction times and congruency). Judgments in congruent conditions (i.e., attractive faces preceded by positive semantic information, or unattractive faces preceded by negative semantic information) are faster than judgments in neutral semantic conditions.

H2 (Reaction times and incongruency). Judgments in incongruent conditions (i.e., attractive faces preceded by negative semantic information, or unattractive faces preceded by positive semantic information) are slower than judgments in neutral semantic conditions.

These hypotheses are operationalized through two dependent measures: the proportion of faces judged as attractive and reaction times. Specifically, a higher proportion of attractive responses is expected for faces preceded by positive words compared to neutral words. In addition, reaction times are expected to be shorter for attractive judgments following positive words, and longer for unattractive judgments following positive words, relative to neutral word conditions.

At the computational level, these effects are expected to be reflected in DDM parameters, with positive semantic valence biasing the starting point of evidence accumulation toward the attractive response and/or increasing the drift rate toward that response.

4 METHODS

4.1 Participants

A total of 53 participants were originally recruited on a voluntary basis for the study, and all provided informed consent prior to participation. The study was conducted in accordance with established ethical guidelines, and

all collected data were anonymized to ensure participant confidentiality and privacy. Basic demographic information was collected, including age, sex, and highest level of education, in order to characterize the sample. Because the task involved reading and processing semantic information in French, participants also reported their self-rated proficiency in French. This measure was used to ensure adequate language comprehension for task performance and to control for variability related to language processing.

Prior to formal analysis, a strict data-cleaning procedure was implemented. First, three participants were excluded due to technical issues during data collection. Second, three lower-proficiency French speakers (categorized at or below the Intermediate B1/B2 level) were excluded to ensure native or highly proficient bilingual processing, given the linguistic demands of the task. No exclusion criteria related to age or gender were applied.

The final sample consisted of 47 participants after exclusion criteria were applied. Participants were aged between 20 and 62 years ($M = 31.36$, $SD = 14.71$; $Mdn = 24$). The sample included 28 men (59.6%), 16 women (34.0%), and 3 participants identifying as another gender or prefer not to say (6.4%). Regarding educational attainment, participants were predominantly highly educated, with 46.8% holding a Master's degree (Bac+5), 21.3% a Bachelor's degree (Bac+3), 17.0% a high school diploma (Bac), 6.4% a Doctoral degree, 6.4% Bac+2 qualifications, and 2.1% a lower secondary education (Brevet). French proficiency was generally high across the sample, with 89.4% native speakers, 6.4% reporting C2-level bilingual proficiency, and 4.3% reporting C1-level advanced proficiency.

From the raw trial matrix generated by these participants, extreme reaction-time outliers were removed using the Tukey (boxplot) method. Specifically, outliers were defined using the interquartile range (IQR) criterion [Enderlein], with lower and upper bounds set at $Q_1 - 1.5 \times IQR$ and $Q_3 + 1.5 \times IQR$, respectively. Any trial falling outside these bounds was excluded from further analysis.

Seven stimuli that were misclassified by more than 45% of participants were removed from the analysis, as they were considered ambiguous. This procedure resulted in a final dataset comprising 2,401 trials for the primary factorial analysis and 1,598 trials for the secondary congruence analysis. Trials with neutral semantic valence were excluded from the congruence analysis because they did not correspond to either congruent or incongruent pairings.

4.2 Stimuli and attractiveness assessment

Facial attractiveness is a subjective evaluative judgment that may vary across observers. As a result, the same face can be perceived as attractive or unattractive depending on individual preferences. To address this variability, several methodological approaches have been proposed in the literature.

One approach consists in using face stimuli that have been pre-rated for attractiveness by an independent sample, allowing faces to be categorized as attractive or unattractive. Alternatively, attractiveness can be assessed by collecting

ratings from a large group of participants on a Likert scale (see Appendix 1), and selecting stimuli based on average ratings, as implemented by [Cui et al.]. A third approach involves collecting individual attractiveness ratings prior to the experimental task, allowing judgments to be tailored to each participant's subjective perception [Zhang et al.].

Given the constraints of this lab project, we opted for the second approach: using a standardized set of stimuli. We selected faces from the Chicago Face Database¹ which provides established norming data based on ratings of facial attractiveness on a scale from 1 (unattractive) to 7 (attractive). This allowed us to categorize stimuli into high (faces with scores above 5) and low (faces with scores below 3) attractiveness groups without the need for a separate pre-rating phase. This method ensures that our semantic valence manipulation is tested against a reliable baseline of facial attractiveness.

For the semantic information, we used the database Emotional Norms [RIC et al., 2013] which contains 524 French personality traits words rated on a 7 point scale from -3 (extremely negative) to +3 (extremely positive). We selected 20 words with the worst scores (below 2.2) to provide negative semantic information (e.g., racist, cruel), 20 words scores between 0.21 and 0.225 for neutral semantic information (e.g., solitary, excited), and 20 words with the highest scores (above 2.2) to provide positive semantic information (e.g., intelligent, marvelous).

4.3 Procedure

Each participant will be tested by a website create by our own. Instructions will be presented on the computer screen, and participants will be informed that they will be asked to judge the attractiveness of faces as quickly as possible.

Participants will be instructed to indicate whether they find the face attractive or unattractive by pressing one of two predefined keys on the keyboard. Reaction times will be measured starting from the moment the face appears on the screen. Different participants will use different keys to indicate whether they find a face attractive or unattractive, in order to avoid any bias related to the choice of keys such as predominant hand bias.

Each trial began with a fixation cross presented at the center of the screen (300 ms), followed by a word with either positive or neutral semantic valence (800 ms). After a 300-ms inter-stimulus interval, a face stimulus was presented at the center of the screen for up to 1500 ms inspired by the methodology of Zhang's team [Zhang et al.].

After a participant responds, the screen will be cleared and the next trial will start. The experiment will include a series of trials where faces of different baseline attractiveness are shown after either a positive or neutral word. The experiment will finish once all trials are completed.

4.4 Experimental design and variables

The experiment follows a within-subjects design. The independent variables are the semantic valence of the word

¹ <https://www.chicagofaces.org/>

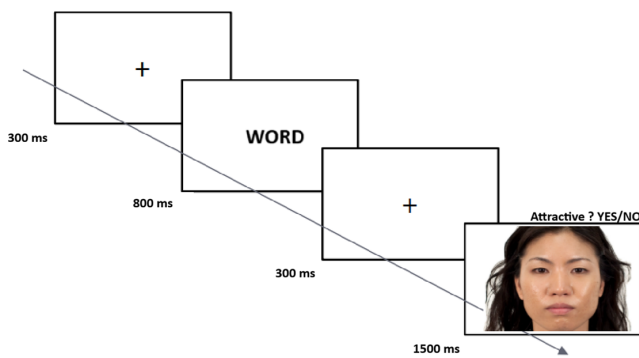


Fig. 2. Schema of the experiment

preceding the face (positive / neutral / negative) and the baseline facial attractiveness of the stimulus (high vs. low). Semantic valence is experimentally manipulated, whereas facial attractiveness is defined based on standardized norming data.

The dependent variables are participants' attractiveness judgments (attractive vs. unattractive) and reaction times measured from the onset of the face stimulus.

At the behavioral level, the hypotheses predict that positive semantic information will increase the likelihood of judging a face as attractive and will modulate reaction times depending on the baseline attractiveness of the face. At the computational level, these effects are expected to be reflected in Diffusion Decision Model parameters, with semantic valence biasing the starting point of evidence accumulation (z) and/or the drift rate (v).

5 RESULTS

5.1 Factorial Analysis: Face Label and Word Valence

A linear mixed-effects model using the Satterthwaite approximation [Satterthwaite] was conducted to examine the effects of face attractiveness category ("Label Face") and preceding semantic word valence on reaction times, with random intercepts included for participants.

The Type III Analysis of Variance revealed a significant main effect of face label, $F(1, 2349.1) = 82.31$, $p < .001$, indicating that reaction times differed depending on whether faces were categorized as attractive or unattractive. No significant main effect of word valence was observed, $F(2, 2349.5) = 0.13$, $p = .878$. However, a significant interaction between face label and semantic word valence was found, $F(2, 2349.3) = 5.45$, $p = .004$, suggesting that the effect of word valence on reaction times depended on the attractiveness category of the face. (see Figure 3 and Table 1)

Decomposition of the fixed-effects parameters provided further insight into the interaction effect. Under the negative-word condition (reference baseline), responses to unattractive faces were significantly faster than responses to attractive faces ($\beta = -0.095$ s, $SE = 0.012$, $p < .001$). Within the attractive-face baseline condition, the introduction of neutral words ($\beta = -0.024$ s, $SE = 0.013$, $p = .066$) or positive

words ($\beta = -0.016$ s, $SE = 0.013$, $p = .221$) did not significantly affect reaction times. (see Table 2)

The significant interaction reflected the fact that the reaction-time advantage for unattractive faces was strongest following negative descriptions and significantly attenuated following neutral ($\beta = 0.056$ s, $SE = 0.017$, $p = .001$) or positive descriptions ($\beta = 0.037$ s, $SE = 0.017$, $p = .033$). The fixed effects explained a modest proportion of variance (Marginal $R^2 = .023$), whereas the full model, including participant-level random effects, accounted for a substantially larger proportion of variance (Conditional $R^2 = .412$). (see Table 2)

5.2 Secondary Structural Analysis: Task Congruence

To examine how semantic expectations interact with perceptual evaluation, a secondary linear mixed-effects model was conducted with trial congruence as the sole fixed predictor ($N = 1,598$ trials). The Type III ANOVA revealed a significant effect of trial congruence, $F(1, 1550.5) = 4.97$, $p = .026$ (see Figure 4 and Table 3).

Relative to the baseline congruent condition ($M = 0.729$ s, $SE = 0.021$), incongruent trials significantly increased reaction times ($\beta = 0.020$ s, $SE = 0.009$, $t(1550) = 2.23$, $p = .026$) (see Table 4). This effect indicates that incongruent semantic-perceptual pairings impose a measurable processing cost.

Pairwise comparisons using the Kenward-Roger approximation confirmed this effect, showing that switching from a congruent to an incongruent condition reliably increased response latency by approximately 0.020 s ($SE = 0.0088$, $t(1551) = 2.23$, $p = .026$).

Although the fixed effect accounted for a small proportion of variance (Marginal $R^2 = .002$), the model explained a substantially larger proportion of variability when accounting for participant-level differences (Conditional $R^2 = .371$), indicating that individual differences strongly contributed to overall reaction-time variability.

To uncover the latent cognitive dynamics underlying the reaction time differences, a computational Drift Diffusion Model (DDM) was evaluated (see Figure 5). The results revealed a dual-stage cognitive cost associated with evaluative incongruence. First, incongruent trials produced greater boundary separation ($a_{\text{congruent}} = 1.61$ vs. $a_{\text{incongruent}} = 1.68$), indicating that participants adopted a more cautious decision policy under visual-semantic conflict. Second, incongruence reduced the magnitude of the drift rate ($v_{\text{congruent}} = -0.70$ vs. $v_{\text{incongruent}} = -0.53$), suggesting less efficient evidence accumulation during decision formation. Together, these parameter shifts indicate that the congruence effect reflects both increased response caution and impaired information processing efficiency. Changes in non-decision time and starting-point bias were comparatively small.

6 DISCUSSION

Analysis of mean reaction times based on preceding semantic valence and facial categorization showed a strong non-linear interaction. Diffusion Decision Modeling (DDM) helped

explain this behavioral asymmetry by showing that the effect was mainly caused by differences in the evidence accumulation process. In both congruent and incongruent conditions, drift rates (v) were consistently directed toward the lower boundary associated with the “Unattractive” response.

Psychologically, participants showed an initial starting-point bias ($z > .50$) toward the upper “Attractive” boundary before making a decision, similarly as in Axt’s study [Axt and Johnson]. This may reflect a social tendency to avoid giving negative judgments about other people’s appearance. However, the visual salience of unattractive faces was strong enough to overcome this initial bias. This explanation is supported by post-experimental feedback, where participants reported using a very strict standard when judging facial attractiveness across the stimulus set. A change in the instruction - by asking to guess the label of the face rather than to judge the facial attractiveness - could have softened this standard.

This overall bias helps explain both the high error rates—which resulted in the exclusion of seven highly ambiguous faces with error rates above 45%—and the strong negative drift rates found in the model. Together, the results suggest that highly unattractive facial features receive priority in processing, quickly reducing the influence of semantic word primes presented beforehand.

We expected the mean reaction time for faces preceded by a neutral semantic word to fall between the reaction times observed in congruent and incongruent conditions. However, the results suggest that there is no true intermediate condition. Instead, participants appeared to process each trial as either a match or a mismatch.

To test this explanation, a follow-up study should be conducted in which participants, similarly to Zhang’s study, first rate a set of faces two weeks before the main experiment. The same face dataset could then be used to examine how semantic information presented before a face influences judgments of facial attractiveness. Another possibility would be to use a pre-labeled face dataset while providing participants with two different sets of instructions, in order to investigate how task instructions affect error rates.

A compelling future avenue for this research, inspired by the TransCog program, involves transitioning from a standard, lossless Diffusion Decision Model to a Leaky Evidence Accumulation Process (LEAP) [Pereira et al.]. The LEAP framework extends traditional sequential sampling models by formalizing three key cognitive dimensions: the continuous temporal decay (leakage) of accumulated evidence, post-decisional information processing, and the computational derivation of subjective confidence [Pereira et al.]. Implementing this model in future iterations of our paradigm would allow for two critical methodological advancements.

First, systematically manipulating the Inter-Stimulus Interval (ISI) between the semantic prime and the facial target would provide a direct empirical window into the leakage rate (λ) of the halo effect, quantifying exactly how fast semantic expectations fade before visual evidence

accumulation begins. Second, integrating an explicit post-decisional confidence scale immediately following the binary attractiveness choice would help elucidate the latent dynamics of evaluative mismatches such as trials where highly consensually attractive faces are rejected due to intense visuo-semantic conflict. This computational approach would fundamentally refine our understanding of how high-level social descriptions and low-level perceptual features compete during rapid, split-second social choices.

6.1 Limitations

Several methodological limitations should be acknowledged when interpreting the findings of the present study. First, due to the inherent time constraints and logistical scope of this lab project, we were unable to conduct an internal, sample-specific pre-rating phase to validate the facial stimuli directly within our own participant pool. Instead, the experimental design relied strictly on external normative data from the Chicago Face Database to categorize the stimuli into high and low attractiveness baselines.

Because facial attractiveness is a deeply subjective evaluative judgment that can vary significantly across observers, this reliance on standardized external baselines introduced substantial idiosyncratic variance. The discrepancy between the database norms and our sample’s actual perceptual consensus resulted in a high degree of stimulus ambiguity for certain faces. This was evidenced by elevated error rates on specific trials, which ultimately necessitated the post-hoc exclusion of seven highly ambiguous face stimuli to maintain the integrity of the primary analyses. This limitation underscores the necessity for future cross-modal priming studies to establish individualized or sample-validated baselines to minimize uncontrolled visuo-semantic conflict.

Furthermore, our paradigm operated under the assumption, supported by foundational meta-analytic evidence [Langlois et al.], that cultural influences on facial attractiveness judgments are largely uniform across populations. However, more recent literature indicates that demographic factors, such as gender and ethnicity, can significantly modulate these evaluative judgments [Coetsee et al., Cro]. By not explicitly controlling for these observer and stimulus characteristics, our study may overlook nuanced variations in how visuo-semantic conflicts unfold across different sub-populations.

6.2 Conclusion

Overall, the results partially support our hypotheses. Congruent semantic–facial pairings produced faster responses than incongruent pairings, supporting H1 and H2 at the congruency level. Importantly, semantic valence did not produce a significant main effect on reaction times, which is consistent with the observed interaction pattern: the influence of semantic information depended on the facial attractiveness category rather than acting uniformly across conditions.

Reaction Times by Word Valence and Face Label

Error bars represent ± 1 Standard Error of the Mean (SEM)

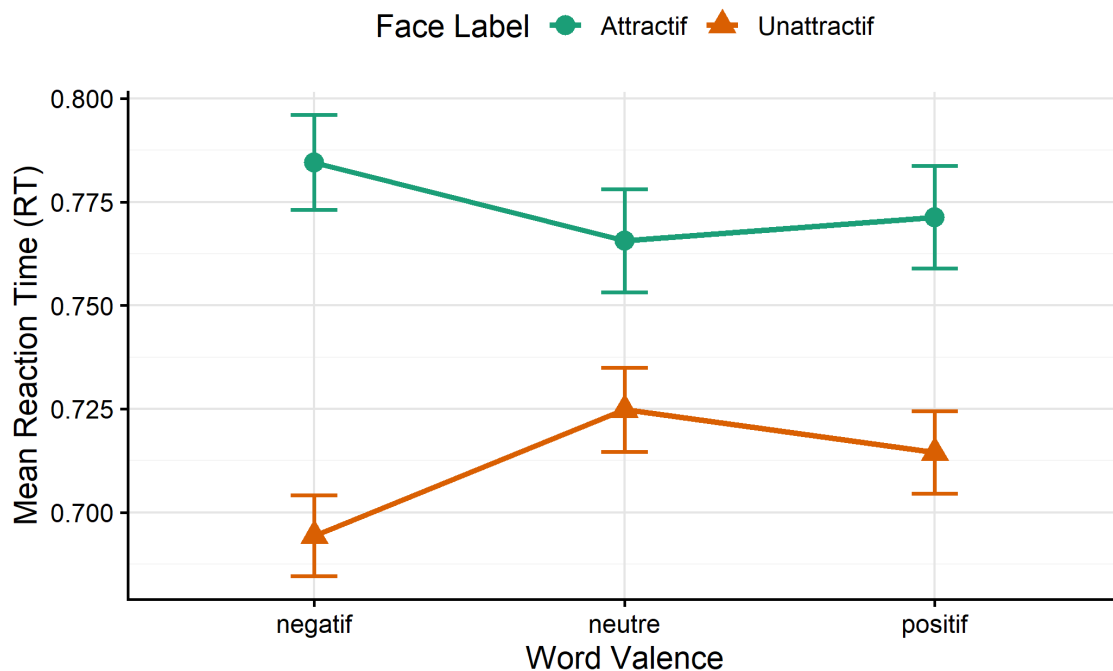


Fig. 3. Mean reaction times (RT) as a function of preceding semantic word valence and target facial attractiveness classification. Green markers represent responses to attractive faces, whereas orange markers represent responses to unattractive faces. Error bars indicate ± 1 Standard Error of the Mean (SEM). The pattern illustrates the significant interaction between face attractiveness classification and semantic word valence ($p = .004$), showing that the reaction-time difference between attractive and unattractive faces varies across negative, neutral, and positive priming conditions.

Reaction Times by Trial Congruence

Error bars represent ± 1 Standard Error of the Mean (SEM)

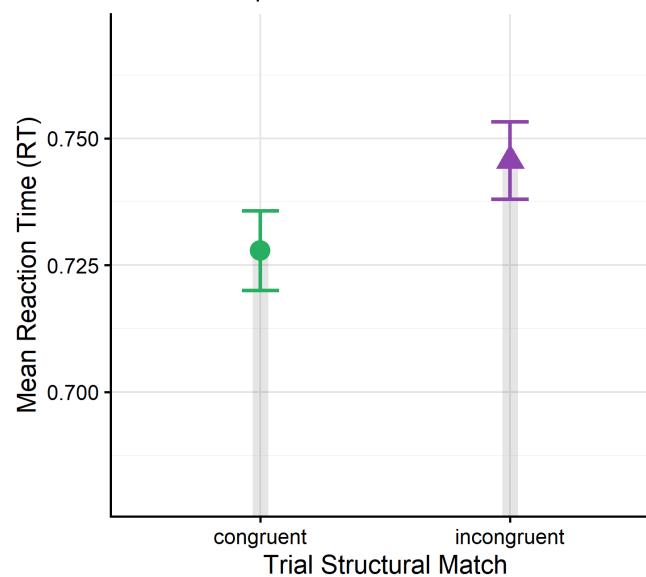


Fig. 4. Reaction time profiles as a function of trial semantic congruence. The green dot displays performance under matched congruent baseline structures, while the purple dot highlights the response latency under emotionally mismatched (incongruent) pairings. Error bars correspond to ± 1 Standard Error of the Mean (SEM). The vertical shift displays the systematic 20 ms processing penalty ($p = .026$) induced by cognitive evaluative conflict.

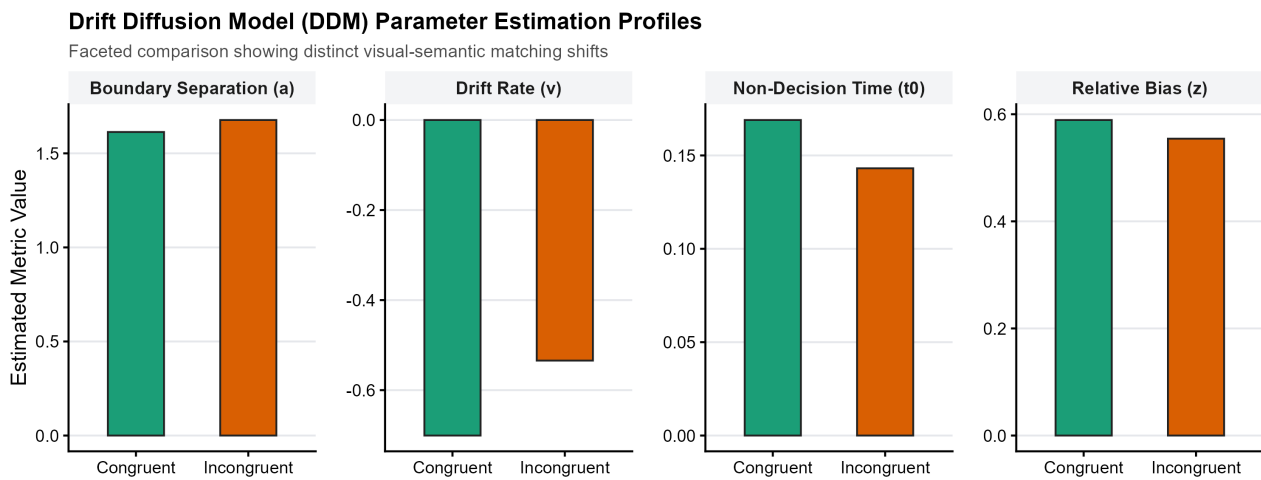


Fig. 5. Faceted Drift Diffusion Model (DDM) parameter estimations across task congruency states. The green bars represent the congruent baseline, and the orange bars represent the incongruent mismatch. Each parameter utilizes an independent, free-floating y-axis scale to isolate absolute structural variations.

Table 1. Type III Analysis of Variance Table via Satterthwaite's Method (Factorial Model)

Predictor	Sum Sq	Mean Sq	NumDF	DenDF	<i>F</i> value	Pr(> <i>F</i>)
Face Label	2.41	2.41	1.00	2349.09	82.31	< .001***
Word Valence	0.01	0.00	2.00	2349.53	0.13	.8778
Face Label × Valence	0.32	0.16	2.00	2349.35	5.45	.0043**

Table 2. Linear Mixed-Effects Model Predicting Reaction Times via Factorial Parameters

Predictor	Estimate (β)	Std. Error	95% CI	df	t value	p
Fixed Effects						
(Intercept) [Ref: Attraktif, Negatif]	0.789	0.022	[0.746, 0.832]	62.02	35.54	< .001***
Face: Unattraktif	-0.095	0.012	[-0.119, -0.071]	2349.23	-7.81	< .001***
Word Valence: Neutre	-0.024	0.013	[-0.049, 0.001]	2349.74	-1.84	.066
Word Valence: Positif	-0.016	0.013	[-0.041, 0.009]	2349.73	-1.22	.221
Unattraktif $\times$ Neutre	0.056	0.017	[0.023, 0.089]	2349.32	3.25	.001**
Unattraktif $\times$ Positif	0.037	0.017	[0.004, 0.070]	2349.37	2.14	.033*
Random Effects & Model Fit						
σ^2 (Residual Variance)	0.029	SD: 0.171				
τ_{00} (Intercept Variance: id)	0.019	SD: 0.139				
Intraclass Correlation (ICC)	0.400	Residual variance clustered within participant ID				
Marginal R^2 / Conditional R^2	$R_m^2 = 0.023, R_c^2 = 0.412$					
AIC / BIC	AIC = -1445.0, BIC = -1398.7					

Note. * $p < .10$, ** $p < .05$, *** $p < .01$, **** $p < .001$. Model compiled over 2,401 active trials across 47 participants.

Table 3. Type III Analysis of Variance Table via Satterthwaite's Method (Congruence Model)

Predictor	Sum Sq	Mean Sq	NumDF	DenDF	<i>F</i> value	Pr(> <i>F</i>)
Congruence	0.15	0.15	1.00	1550.45	4.97	.0259*

Note. * $p < .05$.

Table 4. Linear Mixed-Effects Parameter Statistics Explaining Performance Congruency

Predictor Variable	Model Summary (Reaction Time)
(Intercept) [Ref: Congruent]	0.729*** (0.021)
Congruence: Incongruent	0.020* (0.009)
τ_{00} (Intercept Variance: <i>id</i>)	0.134
σ^2 (Residual Variance: Observations)	0.175
Number of Observations	1598
Marginal R^2	0.002
Conditional R^2	0.371
AIC	-871.9
BIC	-850.4
Intraclass Correlation (ICC)	0.400
RMSE	0.170

Note. * $p < .05$, ** $p < .01$, *** $p < .001$. Standard errors are displayed in parentheses.

DECLARATION OF AI USE

During the preparation of this work, the authors used generative AI tools to assist with manuscript language refinement, Python/Pandas code debugging and optimization, and document layout formatting. After using these services, the authors reviewed, edited, and validated the final output, and take full responsibility for the scientific content and integrity of the report.

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Appendices

1 LIKERT SCALE

The Likert Scale is a psychometric scale named after its inventor, social psychologist Rensis Likert, which is commonly used in research questionnaires. The format of a Likert scale is typically composed of five or seven levels. The seven level scale is:

- 1. Strongly disagree
- 2. Disagree
- 3. Slightly disagree
- 4. Unsure
- 5. Slightly agree
- 6. Agree
- 7. Strongly agree

	Tout à fait d'accord	D'accord	Plutôt d'accord	Indifférent	Plutôt pas d'accord	Pas d'accord	Pas du tout d'accord
je pourrais me passer de mon smartphone durant plusieurs jours	☐	☒	☐	☐	☐	☐	☐
Grâce aux applications de mon smartphone, je ne suis jamais seul	☐	☐	☒	☐	☐	☐	☐
Lorsque je m'ennuie, je planote sur mon smartphone	☐	☐	☐	☐	☒	☐	☐
Je suis dépendant de mon smartphone	☒	☐	☐	☐	☐	☐	☐

Fig. 6. Likert Scale of 7 points from [Benedetto et al.]